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INITIATOR HEAD WITH CIRCUIT BOARD

Abstract

An initiator head may include a housing extending in an axial direction and a circuit board provided in an interior space of the housing. A thickness direction of the circuit board may be substantially parallel with the axial direction. The initiator head may further include a line-in terminal provided on a first side of the housing in the axial direction and accessible from an exterior of the housing. The line-in terminal may be in electrical communication with the circuit board.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] This application is a continuation of U.S. application Ser. No. 18/590,252 filed Feb. 28, 2024, which is a continuation application of U.S. application Ser. No. 17/783,065 filed Jun. 7, 2022, which is a national stage entry of International Application No. PCT/EP2020/085622 filed Dec. 10, 2020, which claims priority to U.S. Provisional Application No. 62/945,942 filed Dec. 10, 2019, U.S. Provisional Application No. 63/001,766 filed Mar. 30, 2020, and U.S. Provisional Application No. 63/003,222, filed Mar. 31, 2020, the contents of each of which are incorporated herein by reference.

BACKGROUND OF THE DISCLOSURE

[0002] Hydrocarbons, such as fossil fuels and natural gas, are extracted from underground wellbores extending deeply below the surface using complex machinery and explosive devices. Once the wellbore is established by placement of cases after drilling, a perforating gun assembly, or train or string of multiple perforating gun assemblies, is lowered into the wellbore and positioned adjacent one or more hydrocarbon reservoirs in underground formations. The perforating gun may have explosive charges which are ignited to create holes in the casing and to blast through the formation so that the hydrocarbons can flow through the casing. Once the perforating gun(s) is properly positioned, a surface signal actuates an ignition of a fuse, which in turn initiates a detonating cord, which detonates the shaped charges to penetrate/perforate the casing and thereby allow formation fluids to flow through the perforations thus formed and into a production string. The surface signal may travel from the surface along electrical wires that run from the surface to one or more initiators, such as ignitors or detonators positioned within the perforating gun assembly.

[0003] Assembly of a perforating gun requires assembly of multiple parts, which may include at least the following components: a housing or outer gun barrel within which is positioned an electrical wire for communicating from the surface to initiate ignition, of an initiator and/or a detonator, a detonating cord, one or more charges and, where necessary, one or more boosters. Assembly may include threaded insertion of one component into another by screwing or twisting the components into place, optionally by use of a tandem adapter. Since the electrical wire must extend through much of the perforating gun assembly, the wire may become easily twisted and crimped during assembly. In addition, when a wired detonator is used it must be manually connected to the electrical wire, which may lead to multiple problems. Due to the rotating assembly of parts, the wires can become torn, twisted and/or crimped/nicked, the wires may be inadvertently disconnected, or even mis-connected in error during assembly. This may lead to costly delays in extracting the hydrocarbons. Additionally, there is a significant safety risk associated with physically and manually wiring live explosives.

[0004] Accordingly, there may be a need for an initiator that would allow for reliable detonation of perforating guns without requiring physically and manually wiring live explosives.

[0005] Additionally, in certain applications, hydraulic fracturing may produce optimal results when perforations are oriented in the direction of maximum principle stress or the preferred fracture

plane (PFP). Perforations oriented in the direction of the PFP create stable perforation tunnels and transverse fractures (perpendicular to the wellbore) that begin at the wellbore face and extend far into the formation. However, if fractures are not oriented in the direction of maximum stress, tortuous, non-transverse fractures may result, creating a complex near-wellbore flow path that can affect the connectivity of the fracture network, increase the chance of premature screen-out, and impede hydrocarbon flow. Accordingly, there may be a need for equipment that can allow for orientation verification of the perforating guns to ensure that perforations are formed in the preferred fracture plane.

BRIEF DESCRIPTION

[0006] An initiator head may include a housing extending in an axial direction and a circuit board provided in an interior space of the housing. A thickness direction of the circuit board may be substantially parallel with the axial direction. The initiator head may further include a line-in terminal provided on a first side of the housing in the axial direction and accessible from an exterior of the housing and a line-out terminal provided on a second side of the housing in the axial direction opposite to the first side. The line-in terminal and the line-out terminal may be in electrical communication with the circuit board.

[0007] An initiator head may include a housing extending in an axial direction and a circuit board provided in an interior space of the housing. A thickness direction of the circuit board may be substantially parallel with the axial direction. The initiator head may further include a line-in terminal provided on a first side of the housing in the axial direction and accessible from an exterior of the housing and a ground terminal provided on a second side of the housing in the axial direction opposite to the first side. The line-in terminal and the ground terminal may be in electrical communication with the circuit board.

[0008] An initiator head may include a housing extending in an axial direction a circuit board provided in an interior space of the housing. A thickness direction of the circuit board may be substantially parallel with the axial direction. The initiator head may further include a line-in terminal provided on a first side of the housing in the axial direction and accessible from an exterior of the housing. The line-in terminal may be in electrical communication with the circuit board.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0009] A more particular description will be rendered by reference to exemplary embodiments that are illustrated in the accompanying figures. Understanding that these drawings depict exemplary embodiments and do not limit the scope of this disclosure, the exemplary embodiments will be described and explained with additional specificity and detail through the use of the accompanying drawings in which:

[0010] FIG. 1 is a cross section view of an initiator head according to an exemplary embodiment;

[0011] FIG. 2 is a perspective view of an initiator according to an exemplary embodiment;

[0012] FIG. 3 is a perspective view of an initiator according to an exemplary embodiment;

[0013] FIG. 4 is a partial, cross section view of an initiator according to an exemplary embodiment, showing a cutaway view of a head and a cross-section of an initiator shell;

[0014] FIG. 5 is a partial cross section view of an initiator according to an exemplary embodiment, showing a cutaway view of a head and a cross-section of an initiator shell;

[0015] FIG. 6 is a partial, cross section view of an initiator, illustrating contents of an initiator shell according to an exemplary embodiment;

[0016] FIG. 7 is a cross section view of an initiator according to an exemplary embodiment;

[0017] FIG. 8 is a perspective view of an initiator engaged with terminals according to an

exemplary embodiment;

[0018] FIG. **9** is a bottom up view of an initiator engaged with terminals according to an exemplary embodiment;

[0019] FIG. **10** is a plan view of an initiator holder and terminals according to an exemplary embodiment;

[0020] FIG. **11** is a plan view of an initiator head and initiator holder according to an exemplary embodiment;

[0021] FIG. **12** is a plan view of an initiator holder and terminals according to an exemplary embodiment;

[0022] FIG. **13** is a plan view of an initiator head an initiator holder according to an exemplary embodiment;

[0023] FIG. **14** is a cutaway perspective view of an initiator head according to an exemplary embodiment;

[0024] FIG. **15** is a perspective view of a fuse connector assembly according to an exemplary embodiment;

[0025] FIG. **16** is a cutaway perspective view of an initiator head according to an exemplary embodiment;

[0026] FIG. **17** is a perspective view of a fuse connector assembly according to an exemplary embodiment;

[0027] FIG. **18** is a cutaway perspective view of an initiator head according to an exemplary embodiment; and

[0028] FIG. **19** is a perspective view of a fuse connector assembly according to an exemplary embodiment.

[0029] Various features, aspects, and advantages of the exemplary embodiments will become more apparent from the following detailed description, along with the accompanying drawings in which like numerals represent like components throughout the figures and detailed description. The various described features are not necessarily drawn to scale in the drawings but are drawn to emphasize specific features relevant to some embodiments.

[0030] The headings used herein are for organizational purposes only and are not meant to limit the scope of the disclosure or the claims. To facilitate understanding, reference numerals have been used, where possible, to designate like elements common to the figures.

DETAILED DESCRIPTION

[0031] Reference will now be made in detail to various embodiments. Each example is provided by way of explanation and is not meant as a limitation and does not constitute a definition of all possible embodiments.

[0032] FIGS. **1-7** show an exemplary embodiment of an initiator head **200**. The initiator head may include a housing **201**, a circuit board **210**, a line-in terminal **212**, a line-out terminal **214**, a ground terminal **216**, a stem **250**, and a fuse **260**.

[0033] As seen in FIG. **1**, the housing **201** may extend in an axial direction **302** and may define an interior space **202**. The housing **201** may be formed of an insulating material, and may be formed by molding, 3D-printing, additive manufacturing, subtractive manufacturing, or any other suitable method. For example, in an exemplary embodiment, the housing **201** may be formed of a non-conductive plastic material such as polyamide. The housing **201** may include a first housing piece **230** and a second housing piece **240** engaged together. Alternatively, the housing **201** may be an integral or monolithic piece molded or additively manufactured around the circuit board **210**.

[0034] FIG. **1** further shows that an exemplary embodiment of the first housing piece **230** may include a first plate **232**. A thickness direction of the first plate **232** may be substantially parallel to the axial direction **302**. As further seen in FIGS. **1-2**, an exemplary embodiment of the first plate **232** may be shaped as an annulus having a substantially circular periphery and a substantially circular through hole **236**. The through hole **236** may be structured to expose the line-in terminal

212 to an exterior **204** of the housing **201**. The first plate **232** may further include a sloped wall **220** sloping from the first plate in the axial direction **302** toward the circuit board **210**. The sloped wall **220** may help to guide a contact pin to contact with the line-in terminal **212**. The first housing piece **230** may further include a first outer peripheral wall **234** extending from the first plate **232** in the axial direction **302**. FIG. 1 and FIG. 4 show an exemplary embodiment in which the first outer peripheral wall **234** extends from an outer periphery of the first plate **232**.

[0035] FIG. 1 further shows that an exemplary embodiment of the second housing piece **240** may include a second plate **242**. A thickness direction of the second plate **242** may be substantially parallel to the axial direction **302**. As further seen in FIG. 3, an exemplary embodiment of the second plate **242** may be substantially circular in shape. The second plate **242** may further include through holes **246** structured to expose the line-out terminal **214** and the ground terminal **216** to an exterior **204** of the housing **201**. The second housing piece **240** may further include a second outer peripheral wall **244** extending from the second plate **242** in the axial direction **302**. FIG. 1 and FIG. 3 show an exemplary embodiment in which the second outer peripheral wall **244** extends from an outer periphery of the second plate **242**.

[0036] As further seen in FIG. 1, the first outer peripheral wall **234** and the second outer peripheral wall **244** may overlap in the axial direction, such that the interior space **202** is formed between the first plate **232** and the second plate **242** in the axial direction. In other words, the interior space **202** may be bounded by the first housing piece **230** and the second housing piece **240**. In an exemplary embodiment, a first housing piece radius of the first housing piece **230** may be smaller than a second housing piece radius of the second housing piece **240**. Thus, the first housing piece **230** may be received within the second housing piece **240** with the first outer peripheral wall **234** being provided between the first plate **232** and the second plate **242** in the axial direction **302**.

Alternatively, the first housing piece radius may be larger than the second housing piece radius, and the second housing piece **240** may be received within the first housing piece **230**, with the second peripheral wall **234** being provided between the first plate **232** and the second plate **242** in the axial direction **302**.

[0037] The first housing piece **230** and the second housing piece **240** may be dimensioned such that the first housing piece **230** and the second housing piece **240** fit snugly together so as not to separate under normal operating conditions. Alternatively, the first housing piece **230** and the second housing piece **240** may be provided with a coupling mechanism such as hook or protrusion and a complementary recess, so that the first housing piece **230** and the second housing piece **240** may snap together. Alternatively, the first outer peripheral wall **234** and the second outer peripheral wall **244** may be complementarily threaded so that the first housing piece **230** and the second housing piece **240** may screw together. Alternatively, the first housing piece **230** and the second housing piece **240** may be bonded together with adhesive.

[0038] FIG. 1 further shows an exemplary embodiment of a circuit board **210**. A thickness direction **211** of the circuit board **210** may be substantially parallel with the axial direction **302**. As explained in further detail herein, orienting the thickness direction **211** substantially parallel with the axial direction **302** allows room for larger firing capacitors and/or surface mounted components **270** to be mounted on the circuit board **210**.

[0039] In an exemplary embodiment, the line-in terminal **212**, the line-out terminal **214**, the ground terminal **216**, and the fuse **260** may be in electrical communication with the circuit board **210**. The line-in terminal **212** may be provided on a first side of the circuit board **210** in the axial direction, and thereby the line-in terminal **212** may be provided on a first side of the housing **201** in the axial direction (i.e., to the left in FIG. 1). The line-out terminal **214** and the ground terminal **216** may be provided on a second side of the circuit board **210** in the axial direction opposite to the first side (i.e., to the right in FIG. 1). The line-out terminal **214** may be configured to output a signal received by the line-in terminal **212**, either directly or in response to processing by the circuit board **210**, as described in detail herein, by being in electrical communication with either the line-in

terminal **212** or the circuit board **210**.

[0040] FIG. **3** shows an exemplary embodiment in which a plurality of line-out terminals **214** and a plurality of ground terminals **216** are provided. The plurality of line-out terminals **214** and the plurality of ground terminals **216** provide a layer of redundancy to help ensure sufficient connection of the initiator head **200** to external electrical components, as explained in detail herein. Each line-out terminal **214** of the plurality of line-out terminals **214** may be directly connected to each other within the housing **201** or on the circuit board **210**. In other words, if one line-out terminal **214** is in electrical communication with the circuit board **210**, then each line-out terminal **214** of the plurality of line-out terminals **214** may be in electrical communication with the circuit board **210**. Similarly, if one line-out terminal **214** becomes in electrical communication with the line-in terminal **212**, then each line-out terminal **214** of the plurality of line-out terminals may be in electrical communication with the line-in terminal **212**. Similarly, if one ground terminal **216** is in electrical communication with the circuit board **210**, then each ground terminal **216** of the plurality of ground terminals **216** may be in electrical communication with the circuit board **210**.

[0041] As further seen in FIG. **1** and FIG. **7**, the circuit board **210** may be a printed circuit board and/or may include one or more surface mounted components **270**. The arrangement of the circuit board **210** and the shape of the initiator head **200** may provide sufficient space in the interior space **202** to accommodate a variety of surface mounted components **270**. In an exemplary embodiment, the surface mounted component **270** of the circuit board **210** may be an integrated circuit (IC) with a dedicated function, a programmable IC, or a microprocessor IC. The circuit board **210** may be configured to activate the fuse **260** in response to a control signal received at the line-in terminal **212**. For example, a user may send a firing signal via a firing panel. The firing signal may be received at the line-in terminal **212**, and the circuit board **210**, through ICs provided on the circuit board **210**, may process the firing signal and activate the fuse **260**. Additionally, the circuit board **210** may include a switch circuit configured to establish electrical communication between the line-out terminal **214** and the line-in terminal **212** in response to a predetermined switch signal. The line-out terminal **214** may be in electrical communication with subsequent initiator heads **200** provided downstream in a string of connected perforating guns, thereby allowing a user to send switch signals to toggle which initiator head is active to receive a firing command.

[0042] In an exemplary embodiment, one of the surface mounted components **270** may be one selected from a group consisting of a temperature sensor, an orientation sensor, a safety circuit, and a capacitor. Readings from one of these components may be used by a microprocessor on circuit board **210** to determine when it is appropriate to activate the fuse **260**. The temperature sensor may be configured to measure temperature of the wellbore environment and provide a signal corresponding to the temperature to the circuit board **210**. The orientation sensor may include, but is not limited to, an accelerometer, a gyroscope, and/or a magnetometer. The orientation sensor may be configured to determine an orientation of the initiator head **200** within the wellbore, which, if the orientation of the initiator head is fixed relative to a charge holder, can be used to determine an orientation of the charge(s) in the perforating gun. In an exemplary embodiment, the orientation sensor may determine an orientation of the initiator head **200** relative to gravity. Alternatively, the orientation sensor may determine an orientation of the initiator head relative an ambient magnetic field. The safety circuit may provide additional safety precautions to prevent unintentional activation of the initiator **100**. The capacitor may be used to store a voltage to activate the fuse **260**. The size of the interior space **202** may allow for a larger capacity capacitor to be used. This allows a larger discharge voltage for activating the fuse **260**, which may help to ensure more reliable activation of the fuse **260**.

[0043] FIG. **1** and FIGS. **4-7** further show an exemplary embodiment of the stem **250**. The stem **250** may extend in the axial direction **302** from the housing **201**. In an exemplary embodiment, the stem **250** may be formed of the same material as the second housing piece **240** and may be integrally and/or monolithically formed with the second plate **242**. Alternatively, the stem may be

formed as a separate piece and mechanically connected to the second housing piece via clips or mated structures such as protrusions and recesses, or adhesively connected using an adhesive.

[0044] As seen in FIG. 1, the stem 250 may include a stem outer peripheral wall 252. The stem outer peripheral wall 252 may define a stem cavity 254 provided radially inward from the stem outer peripheral wall 252. A first discharge channel 256 and a second discharge channel 258 may connect the stem cavity 254 and the interior space 202 of the housing 201. The first discharge channel 256 may accommodate therein a first discharge terminal 218 in electrical communication with the circuit board 210. In other words, the first discharge terminal 218 may extend from the circuit board 210 into the first discharge channel 256. Similarly, the second discharge channel 256 may accommodate therein a second discharge terminal 219 in electrical communication with the circuit board 210. In other words, the second discharge terminal 219 may extend from the circuit board 210 into the second discharge channel 258.

[0045] FIG. 1 further shows that, in an exemplary embodiment, the fuse 260 may be provided within the stem cavity 254. A first end of a first fuse terminal 262 may be in electrical communication with the first discharge terminal 218 within the first discharge channel 256, and a second end of the first fuse terminal may be proximate to the fuse 260. A first end of a second fuse terminal 264 may be in electrical communication with the second discharge terminal 219 within the second discharge channel 258, and a second end of the second fuse terminal 264 may be proximate to the fuse 260 and the second end of the first fuse terminal 262. The circuit board 210 may be configured to activate the fuse 260 in response to a control signal by discharging a stored voltage across the first fuse terminal 262 and the second fuse terminal 264. The store voltage may be stored in a capacitor in electrical communication with the circuit board 210. In an exemplary embodiment, the capacitor may be one of the surface mounted components 270 provided on the circuit board 210. The proximity of the second end of the first fuse terminal 262 and the second end of the second fuse terminal 264 may allow for the generation of a spark when the stored voltage is discharged, thereby activating the fuse 260. In an exemplary embodiment, activating the fuse 260 may include igniting or detonating the fuse 260.

[0046] As seen in FIG. 6, an exemplary embodiment of the stem 250 may include a window 253 cut through the stem outer peripheral wall 252. The window 253 may allow access for a user to connect the first discharge terminal 218 to the first fuse terminal 262 and the second discharge terminal 219 to the second fuse terminal 264, such as by soldering, during assembly of the initiator head 200.

[0047] FIGS. 14-19 show exemplary embodiments in which the circuit board 210 is in electrical communication with the fuse 260 via direct physical contact, so as to streamline the manufacturing process by eliminating soldering between the circuit board 210 and the fuse 260. For example, FIG. 14 shows an exemplary embodiment in which the circuit board 210 is in electrical communication with the fuse 260 via a fuse connector assembly 600. The fuse connector assembly 600 may include a first discharge connector 602 configured to receive and make direct electrical contact with the first fuse terminal 262 and a second discharge connector 604 configured to receive and make direct electrical contact with the second fuse terminal 264 (not shown in FIG. 14).

[0048] The fuse connector assembly 600 may include a mounting block 606, the first discharge connector 602 extending through the mounting block 606, and the second discharge connector 604 extending through the mounting block 606. The mounting block 606 may be formed of an insulating material and may facilitate connection and/or fastening of the fuse connector assembly 600 to the circuit board 210. Further, the mounting block 606 may provide mechanical strength and support for the fuse connector assembly 600. When the fuse connector assembly 600 is connected to the circuit board 210, the first discharge connector 602 and the second discharge connector 604 may extend from the circuit board 210 into the stem 250.

[0049] FIG. 15 further shows an exemplary embodiment of the first discharge connector 602. For simplicity, only the first discharge connector 602 is described in detail herein; it will be understood

from FIG. 15 that the second discharge connector **604** may be substantially similar to the first discharge connector **602** in terms of structure. The first discharge connector **602** may be formed of an electrically conductive material. The first discharge connector **602** may include a first body portion **610**, and a first board connector terminal **612** may be provided at a first end of the first body portion **610**. The first board connector terminal **612** may connect to the circuit board **210**. [0050] The first discharge connector **602** may further include a first base portion **620** and a second base portion **630** extending from the first body portion **610** at a second end of the first body portion **610**. The first discharge connector **602** may further include a first arm portion **622** extending from the first base portion **620** and a second arm portion **632** extending from the second base portion **630**. The first arm portion **622** may be bent or inclined in a direction toward the second arm portion **632**. Similarly, the second arm portion **632** may be bent or inclined in a direction toward the first arm portion **622**. The first discharge connector **602** may further include a first tip portion **624** at an end of the first arm portion **622** and a second tip portion **634** at an end of the second arm portion **632**. The first tip portion **624** may be bent or inclined in a direction away from the second tip portion **634**. Similarly, the second tip portion **634** may be bent or inclined in a direction away from the first tip portion **624**.

[0051] A first contact portion **626** may be formed between the first arm portion **622** and the first tip portion **624**, and a second contact portion **636** may be formed between the second arm portion **632** and the second tip portion **634**. The first contact portion **626** may be resiliently biased toward the second contact portion **636** based on the connection between the first base portion **620** and the first arm portion **622**. Similarly, the second contact portion **636** may be resiliently biased toward the first contact portion **626** based on the connection between the second base portion **630** and the second arm portion **632**. The first contact portion **626** may be in contact with the second contact portion **636**. Alternatively, there may be a gap between the first contact portion **626** and the second contact portion **636**. In an exemplary embodiment, a size of the gap may be less than a thickness of the first fuse terminal **262**.

[0052] The first discharge connector **602** may be configured to receive, and make electrical contact with, the first fuse terminal **262**. Similarly, the second discharge connector **604** may be configured to receive, and make electrical contact with, the second fuse terminal **264**. For example, during assembly of the initiator head **200**, the circuit board **210** and the fuse **260** may be pushed together in the axial direction **302**, thereby bringing the first fuse terminal **262** into contact with the first tip portion **624** and the second tip portion **634**. Further relative motion between the fuse **260** and the circuit board **210** may cause the first fuse terminal **262** to deflect the first tip portion **624** and the second tip portion **634** away from each other. The first fuse terminal **262** may then be in contact with the first contact portion **626** and the second contact portion **636**, i.e., sandwiched between the first contact portion **626** and the second contact portion **636**. The resilient bias of the first contact portion **626** and the second contact portion **636** may help to maintain contact, and thus electrical communication, between the first contact portion **626**, the second contact portion **636**, and the first fuse terminal **262**. It will be understood that contact between the second discharge connector **604** and the second fuse terminal **264** may be achieved in a similar way. The window **253** may allow for visual confirmation of the connection between the first discharge connector **602** and the first fuse terminal **262** and between the second discharge connector **604** and the second fuse terminal **264**.

[0053] FIG. 16 shows an exemplary embodiment in which the circuit board **210** is in electrical communication with the fuse **260** via a fuse connector assembly **700**. The fuse connector assembly **700** may include a first discharge connector **702** configured to receive and make direct electrical contact with the first fuse terminal **262** and a second discharge connector **704** configured to receive and make direct electrical contact with the second fuse terminal **264** (not shown in FIG. 16).

[0054] The fuse connector assembly **700** may include a mounting block **706**, the first discharge connector **702** extending through the mounting block **706**, and the second discharge connector **704** extending through the mounting block **706**. The mounting block **706** may be formed of an

insulating material and may facilitate connection and/or fastening of the fuse connector assembly **700** to the circuit board **210**. Further, the mounting block **706** may provide mechanical strength and support for the fuse connector assembly **700**. When the fuse connector assembly **700** is connected to the circuit board **210**, the first discharge connector **702** and the second discharge connector **704** may extend from the circuit board **210** into the stem **250**.

[0055] FIG. **17** further shows an exemplary embodiment of the first discharge connector **702**. For simplicity, only the first discharge connector **702** is described in detail herein; it will be understood from FIG. **17** that the second discharge connector **704** may be substantially similar to the first discharge connector **702** in terms of structure. The first discharge connector **702** may be formed of an electrically conductive material. The first discharge connector **702** may include a first body portion **710**, and a first board connector terminal **712** may be provided at a first end of the first body portion **710**. The first board connector terminal **712** may connect to the circuit board **210**.

[0056] The first discharge connector **702** may further include a first base portion **720** and a second base portion **730** extending from the first body portion **710** at a second end of the first body portion **710**. The first discharge connector **702** may further include a first arm portion **722** extending from the first base portion **720** and a second arm portion **732** extending from the second base portion **730**. The first arm portion **722** may be bent or inclined in a direction away from the second arm portion **732**. Similarly, the second arm portion **732** may be bent or inclined in a direction away from the first arm portion **722**. The first discharge connector **702** may further include a first tip portion **724** at an end of the first arm portion **722** and a second tip portion **734** at an end of the second arm portion **732**. The first tip portion **724** may be bent or inclined in a direction toward the second tip portion **734** and back toward the first body portion **710**. Similarly, the second tip portion **734** may be bent or inclined in a direction toward the first tip portion **724** and back toward the first body portion **710**.

[0057] A first contact portion **726** may be formed at an end of the first tip portion **724**, and a second contact portion **736** may be formed at an end of the second tip portion **734**. The first contact portion **726** may be resiliently biased toward the second contact portion **736** based on the connection between the first base portion **720** and the first arm portion **722**. Similarly, the second contact portion **736** may be resiliently biased toward the first contact portion **726** based on the connection between the second base portion **730** and the second arm portion **732**. The first contact portion **726** may be in contact with the second contact portion **736**. Alternatively, there may be a gap between the first contact portion **726** and the second contact portion **736**. In an exemplary embodiment, a size of the gap may be less than a thickness of the first fuse terminal **262**.

[0058] The first discharge connector **702** may be configured to receive, and make electrical contact with, the first fuse terminal **262**. Similarly, the second discharge connector **704** may be configured to receive, and make electrical contact with, the second fuse terminal **264**. For example, during assembly of the initiator head **200**, the circuit board **210** and the fuse **260** may be pushed together in the axial direction **302**, thereby bringing the first fuse terminal **262** into contact with the first tip portion **724** and the second tip portion **734**. Further relative motion between the fuse **260** and the circuit board **210** may cause the first fuse terminal **262** to deflect the first tip portion **724** and the second tip portion **734** away from each other. The first fuse terminal **262** may then be in contact with the first contact portion **726** and the second contact portion **736**, i.e., sandwiched between the first contact portion **726** and the second contact portion **736**. The resilient bias of the first contact portion **726** and the second contact portion **736** may help to maintain contact, and thus electrical communication, between the first contact portion **726**, the second contact portion **736**, and the first fuse terminal **262**. It will be understood that contact between the second discharge connector **704** and the second fuse terminal **264** may be achieved in a similar way. The window **253** may allow for visual confirmation of the connection between the first discharge connector **702** and the first fuse terminal **262** and between the second discharge connector **704** and the second fuse terminal **264**.

[0059] FIGS. **18-19** show an exemplary embodiment in which the circuit board **210** is in electrical

communication with the fuse **260** via a fuse connector assembly **800**. The fuse connector assembly **800** is similar in many aspects to the fuse connector assembly **700**; similar structures will be indicated with the same reference numerals, and detailed descriptions of these similar structures will be omitted. In the fuse connector assembly **800**, the first arm portion **822** may include a first arm part **822a** extending from the first base portion **720** and a second arm part **822b** extending from the first arm part **822a**. The second arm portion **832** may include a third arm part **832a** extending from the first base portion **730** and a fourth arm part **832b** extending from the first arm part **832a**. Each of the first arm part **822a** and the third arm part **832a** may be bent or inclined in a direction away from each other. Each of the second arm part **822b** and the fourth arm part **832b** may be bent or inclined in a direction toward each other.

[0060] FIGS. 2-7 shows an exemplary embodiment of an initiator **100**. The initiator **100** may include an initiator head **200** and an initiator shell **300**. The initiator head **200** may be similar in structure and function as described in detail above. The initiator shell **300** may be coaxial with the initiator head **200**. In an exemplary embodiment, a head dimension **X1** of the head **200** in a first direction perpendicular to the axial direction **302** may be larger than a shell dimension **X2** in the first direction. According to an aspect, the initiator may be configured as an ignitor or a detonator, depending on the needs of the application.

[0061] In an exemplary embodiment, the initiator shell **300** may include a shell wall **310** and a shell crimp **312** crimped around the stem **250**. The shell wall **310** may extend in the axial direction **302** and may be formed of a deep-drawn metal. Non-limiting examples of the metal used for the shell wall **310** may include aluminum, copper, steel, tin, or brass. Plastics may also be used as a material for the shell wall **310**. The shell wall **310** may define a shell interior **320**. A primary explosive **322** may be provided within the shell interior **320**. In an exemplary embodiment, the circuit board **210** may be configured to activate the primary explosive **322**, and in some embodiments the primary explosive **322** and the secondary explosive **324**, in response to a control signal received at the line-in terminal **212**. For example, the primary explosive **322** may be arranged such that the fuse **260** is within an operable distance of the primary explosive **322**. Being within an operable distance means that the fuse **260** is provided close enough to the primary explosive **322** that the primary explosive **322** is ignited and/or detonated when the fuse **260** is activated. In other words, by activating the fuse **260** in response to a control signal, the circuit board **210** may activate the primary explosive **322**.

[0062] The secondary explosive **324** may abut the primary explosive **322** and seal the primary explosive **322** within a non-mass explosive (NME) body **330**. The primary explosive **322** and the secondary explosive **324** may have a total thickness of about 3 mm to about 30 mm in an exemplary embodiment. Alternatively, the total thickness may be about 3 mm to about 10 mm. The secondary explosive **324** may be configured as a layer of an explosive material. According to an exemplary embodiment, the primary explosive **322** may include at least one of lead azide, silver azide, lead styphnate, tetracene, nitrocellulose, BAX, and a lead azide free primary explosive as described in USPGP 2019/0256438, herein incorporated by reference.

[0063] Each of the primary explosive **322** and the secondary explosive **324** may have a safe temperature rating of above 150° C. (with the exception of PETN, which has a rating of approximately 120° C.). The secondary explosive **324** may include a material that is less sensitive to initiation, as compared to the primary explosive **322**. The secondary explosive **324** may include at least one of PETN, RDX, HMX, HNS and PYX. In an embodiment, the secondary explosive **324** may be less sensitive to initiation than PETN.

[0064] The primary explosive **322** and the secondary explosive **324** may be provided within the NME body **330**. The NME body **330** may help to avoid an unintentional initiation of the primary explosive **322** or the main load explosive **332** by an external mechanical force. The NME body **330** may be composed of an electrically conductive, electrically dissipative or electrostatic discharge (ESD) safe synthetic material. According to an exemplary embodiment, the non-mass-explosive

body **330** may be formed of a metal, such as cast-iron, zinc, machinable steel or aluminum. Alternatively, the NME body **330** may be formed from a plastic material. While the NME body **330** may be made using various processes, the selected process utilized for making the NME body **330** is based, at least in part, by the type of material from which it is made. For instance, when the NME body **330** is made from a plastic material, the selected process may include an injection molding process. When the NME body **330** is made from a metallic material, the NME body **330** may be formed using any conventional CNC machining or metal casting processes.

[0065] The initiator shell **300** may further include a main load explosive **332** provided adjacent the primary explosive **322**, and in embodiment including a secondary explosive **324**, adjacent the secondary explosive **324**. The main load explosive **332** includes compressed secondary explosive materials. According to an aspect, the main load explosive **332** may include one or more of cyclotrimethylenetrinitramine (RDX), octogen/cyclotetramethylenetetranitramine (HMX), hexanitrostilbene (HNS), pentaerythritol tetranitrate (PETN), 2,6-Bis(picrylamino)-3,5-dinitropyridine (PYX), and 1,3,5-triaminio-2,4,6-trinitrobenzene (TATB). The type of explosive material used may be based at least in part on the operational conditions in the wellbore and the temperature downhole to which the explosive may be exposed.

[0066] In an exemplary embodiment shown in FIGS. **11-13**, an exterior shape of the housing **201** may be rotationally asymmetric with respect to the axial direction **302**. In other words, when looking along the axial direction **302**, a periphery of the housing **201** may be shaped such that an orientation of the housing **201** is unique for each angle around the axial direction. For example, FIG. **11** shows that a key protrusion **290** or a key protrusion **292** may be formed on a periphery of the housing **201**, and FIG. **13** shows that a key recess **294** may be formed on a periphery of the housing **201**. As is clear from FIG. **11** and FIG. **13**, there are no possible rotations of the housing **201** where the housing **201** has a matching profile. In other words, an exterior profile of housing **201** is unique for each possible rotation angle. It will be understood that the size, shape, and/or number of key protrusions and/or key recesses is not limited to what is shown in FIG. **11** and FIG. **13**, as long as they create a rotational asymmetry in the shape of housing **201**. Additionally, key protrusions and key recesses may be combined together on a single housing **201**.

[0067] FIGS. **8-13** illustrate an exemplary embodiments of an initiator system **500**. The initiator system **500** may include an initiator holder **400** (see FIGS. **10-13**) and an initiator **100** received within the initiator holder **400**.

[0068] As seen in FIGS. **8-10**, an exemplary embodiment of the initiator holder **400** may include a holder ground terminal **410**. The holder ground terminal **410** may include a holder ground contact **412**. In an exemplary embodiment shown in FIGS. **8-9**, the holder ground contact **412** may be punched from the material of the holder ground terminal **410** and then bent to a side of the holder ground terminal **410**. This may help to impart a spring-loaded action to the holder ground contact **412** and bias the holder ground contact **412** in a direction toward the initiator head **200**, thereby helping to ensure a more secure electrical contact between the ground terminal **216** and the holder ground contact **412**. In other words, when the initiator **100** is positioned within the initiator holder **400**, the holder ground contact **412** may be in electrical communication with the ground terminal **216** (see FIG. **9**) via contact.

[0069] FIGS. **8-10**, and FIG. **12** show that, in an exemplary embodiment of the holder ground terminal **410**, the holder ground contact **412** may be one of a plurality of holder ground contacts **412**. As seen in FIG. **9**, if the initiator head **200** includes a plurality of ground terminals **216**, then the plurality of holder ground contacts **412** provided a layer of redundancy for establishing a connection to ground. For example, even if one pair the ground terminals **216** and the holder ground contacts **412** fails to establish a secure electrical connection, a second pair of the ground terminals **216** and the holder ground contacts **412** may form a secure electrical connection.

[0070] As further seen in FIGS. **10-13**, the initiator holder **400** may further include a holder ground bar **414** extending from the holder ground terminal **410**. The holder ground bar **414** may contact a

ground when the initiator holder **400** is received within a perforating gun. In other words, the holder ground terminal **410** may be in electrical communication with ground, for example through the holder ground bar **414**.

[0071] As further seen in the exemplary embodiment of FIG. **10**, the initiator holder **400** may include a through-wire terminal **420**. The through-wire terminal **420** may include a through-wire contact **422**. In an exemplary embodiment shown in FIGS. **8-9**, the through wire contact **422** may be punched from the material of the through-wire terminal **420** and then bent to a side of the through-wire terminal **420**. This may help to impart a spring-loaded action to the through-wire contact **422** and bias the through-wire contact **422** in a direction toward the initiator head **200**, thereby helping to ensure a more secure electrical contact between the through-wire terminal **214** and the through-wire contact **414**. In other words, when the initiator **100** is positioned within the initiator holder **400**, the through-wire contact **422** may be in electrical communication with the through-wire terminal **214** via contact.

[0072] FIGS. **8-9**, FIG. **10**, and FIG. **12** show that, in an exemplary embodiment of the through-wire terminal **420**, the through-wire contact **422** may be one of a plurality of through-wire contacts **422**. As seen in FIG. **9**, if the initiator head **200** includes a plurality of through-wire terminals **214**, then the plurality of through-wire contacts **422** provided a layer of redundancy for establishing an electrical connection. For example, even if one pair of the through-wire terminals **214** and the through-wire contacts **422** fails to establish a secure electrical connection, a second pair of the through-wire terminals **214** and the through-wire contacts **412** may form a secure electrical connection.

[0073] FIGS. **10-13** show exemplary embodiments of an initiator system **500** comprising a key system configured to ensure a correct alignment between the initiator **100** and the initiator holder **400**. For example, when an initiator **100** is received into holder hole **402**, the initiator **100** may rotate around the axial direction **302**. This could create a misalignment between the through-line terminal(s) **214** and the ground terminal(s) **216** of the initiator head **200** and the through-line contact(s) **422** and holder ground contact(s) **412** of the holder **400**. Accordingly, a key system may be configured to rotationally fix the initiator head **200** relative to the holder **400**, thereby helping to ensure a correct alignment between the initiator **100** and the initiator **400**. In this context, a correct alignment may be an alignment in which the through-line terminal(s) **214** and the ground terminal(s) **216** of the initiator head **200** are correspondingly aligned with the through-line contact(s) **422** and holder ground contact(s) **412** of the holder **400**.

[0074] FIGS. **10-11** show an exemplary embodiment in which recesses **440**, **442** may be formed in an outer peripheral wall **430** of the holder **400**. For example, a first holder recess **440** may be formed partially through the outer peripheral wall **430**. Alternatively or additionally, a second holder recess **442** may be formed through the entire thickness of the outer peripheral wall **430**. As seen in FIG. **11**, an exemplary embodiment of the housing **201** of the initiator head **200** may include a first key protrusion **290** formed on an outer periphery of housing **201**. The first key protrusion **290** may be shaped and sized to fit within the first holder recess **440**. Alternatively or additionally, a second key protrusion **292** may be formed on an outer periphery of the housing **201**. The second key protrusion **292** may be shaped and sized to fit within the second holder recess **442**.

[0075] FIGS. **12-13** show an exemplary embodiment in which protrusions may be formed in the outer peripheral wall **430** of the holder **400**. For example, a holder protrusion **444** may extend radially inwardly from the outer peripheral wall **430**. As seen in FIG. **13**, an exemplary embodiment of the housing **201** of the initiator head **200** may include a housing recess **294** corresponding to the holder protrusion **444**.

[0076] It will be understood from the exemplary embodiments shown in FIGS. **10-13** that the number, size, and shape of recesses and protrusions may be varied to achieve the same effect, as long as the recesses and their corresponding protrusions are rotationally asymmetric around the longitudinal axis. For example, a single recess and a single protrusion may be sufficient to achieve

rotational asymmetry. Alternatively, a plurality of recesses of corresponding protrusions may be used. Further, it will be understood that recesses and protrusions may be mixed on a single piece. For example, an exemplary embodiment of the housing **201** may include both a protrusion and a recess, corresponding to a complementary recess and protrusion on the initiator holder **400**.

[0077] This disclosure, in various embodiments, configurations and aspects, includes components, methods, processes, systems, and/or apparatuses as depicted and described herein, including various embodiments, sub-combinations, and subsets thereof. This disclosure contemplates, in various embodiments, configurations and aspects, the actual or optional use or inclusion of, e.g., components or processes as may be well-known or understood in the art and consistent with this disclosure though not depicted and/or described herein.

[0078] The phrases “at least one,” “one or more,” and “and/or” are open-ended expressions that are both conjunctive and disjunctive in operation. For example, each of the expressions “at least one of A, B and C,” “at least one of A, B, or C,” “one or more of A, B, and C,” “one or more of A, B, or C,” and “A, B, and/or C” means A alone, B alone, C alone, A and B together, A and C together, B and C together, or A, B, and C together.

[0079] In this specification and the claims that follow, reference will be made to a number of terms that have the following meanings. The terms “a” (or “an”) and “the” refer to one or more of that entity, thereby including plural referents unless the context clearly dictates otherwise. As such, the terms “a” (or “an”), “one or more” and “at least one” can be used interchangeably herein.

Furthermore, references to “one embodiment”, “some embodiments”, “an embodiment” and the like are not intended to be interpreted as excluding the existence of additional embodiments that also incorporate the recited features. Approximating language, as used herein throughout the specification and claims, may be applied to modify any quantitative representation that could permissibly vary without resulting in a change in the basic function to which it is related.

Accordingly, a value modified by a term such as “about” is not to be limited to the precise value specified. In some instances, the approximating language may correspond to the precision of an instrument for measuring the value. Terms such as “first,” “second,” “upper,” “lower,” etc. are used to identify one element from another, and unless otherwise specified are not meant to refer to a particular order or number of elements.

[0080] As used herein, the terms “may” and “may be” indicate a possibility of an occurrence within a set of circumstances; a possession of a specified property, characteristic or function; and/or qualify another verb by expressing one or more of an ability, capability, or possibility associated with the qualified verb. Accordingly, usage of “may” and “may be” indicates that a modified term is apparently appropriate, capable, or suitable for an indicated capacity, function, or usage, while taking into account that in some circumstances the modified term may sometimes not be appropriate, capable, or suitable. For example, in some circumstances an event or capacity can be expected, while in other circumstances the event or capacity cannot occur—this distinction is captured by the terms “may” and “may be.”

[0081] As used in the claims, the word “comprises” and its grammatical variants logically also subtend and include phrases of varying and differing extent such as for example, but not limited thereto, “consisting essentially of” and “consisting of.” Where necessary, ranges have been supplied, and those ranges are inclusive of all sub-ranges therebetween. It is to be expected that the appended claims should cover variations in the ranges except where this disclosure makes clear the use of a particular range in certain embodiments.

[0082] The terms “determine,” “calculate,” and “compute,” and variations thereof, as used herein, are used interchangeably and include any type of methodology, process, mathematical operation or technique.

[0083] This disclosure is presented for purposes of illustration and description. This disclosure is not limited to the form or forms disclosed herein. In the Detailed Description of this disclosure, for example, various features of some exemplary embodiments are grouped together to

representatively describe those and other contemplated embodiments, configurations, and aspects, to the extent that including in this disclosure a description of every potential embodiment, variant, and combination of features is not feasible. Thus, the features of the disclosed embodiments, configurations, and aspects may be combined in alternate embodiments, configurations, and aspects not expressly discussed above. For example, the features recited in the following claims lie in less than all features of a single disclosed embodiment, configuration, or aspect. Thus, the following claims are hereby incorporated into this Detailed Description, with each claim standing on its own as a separate embodiment of this disclosure.

[0084] Advances in science and technology may provide variations that are not necessarily express in the terminology of this disclosure although the claims would not necessarily exclude these variations.

Claims

1. An initiator head comprising: a housing extending in an axial direction; a circuit board provided in an interior space of the housing, a thickness direction of the circuit board being substantially parallel with the axial direction; a line-in terminal provided on a first side of the housing in the axial direction and accessible from an exterior of the housing, the line-in terminal being in electrical communication with the circuit board; and a line-out terminal provided on a second side of the housing in the axial direction opposite to the first side, the line-out terminal being in electrical communication with the circuit board.
2. The initiator head of claim 1, wherein: the line-out terminal is one of a plurality of line-out terminals; and each line-out terminal of the plurality of line-out terminals is in electrical communication with the circuit board.
3. The initiator head of claim 1, further comprising: a ground terminal provided on a second side of the housing in the axial direction opposite to the first side; wherein the ground terminal is in electrical communication with the circuit board.
4. The initiator head of claim 3, wherein: the ground terminal is one of a plurality of ground terminals; and each ground terminal of the plurality of ground terminals is in electrical communication with the circuit board.
5. The initiator head of claim 1, wherein an exterior shape of the housing is rotationally asymmetric with respect to the axial direction.
6. The initiator head of claim 1, further comprising a surface mounted component mounted on the circuit board.
7. An initiator head comprising: a housing extending in an axial direction; a circuit board provided in an interior space of the housing, a thickness direction of the circuit board being substantially parallel with the axial direction; a line-in terminal provided on a first side of the housing in the axial direction and accessible from an exterior of the housing, the line-in terminal being in electrical communication with the circuit board; and a ground terminal provided on a second side of the housing in the axial direction opposite to the first side, the ground terminal being in electrical communication with the circuit board.
8. The initiator head of claim 1, wherein: the ground terminal is one of a plurality of ground terminals; and each ground terminal of the plurality of ground terminals is in electrical communication with the circuit board.
9. The initiator head of claim 1, further comprising: a line-out terminal provided on a second side of the housing in the axial direction opposite to the first side; wherein the line-out terminal is in electrical communication with the circuit board.
10. The initiator head of claim 9, wherein: the line-out terminal is one of a plurality of line-out terminals; and each line-out terminal of the plurality of line-out terminals is in electrical communication with the circuit board.

- 11.** The initiator head of claim 1, wherein an exterior shape of the housing is rotationally asymmetric with respect to the axial direction.
- 12.** The initiator head of claim 1, further comprising a surface mounted component mounted on the circuit board.
- 13.** An initiator head comprising: a housing extending in an axial direction; a circuit board provided in an interior space of the housing, a thickness direction of the circuit board being substantially parallel with the axial direction; and a line-in terminal provided on a first side of the housing in the axial direction and accessible from an exterior of the housing, the line-in terminal being in electrical communication with the circuit board.
- 14.** The initiator head of claim 13, further comprising: a line-out terminal provided on a second side of the housing in the axial direction opposite to the first side; wherein the line-out terminal is in electrical communication with the circuit board.
- 15.** The initiator head of claim 14, wherein: the line-out terminal is one of a plurality of line-out terminals; and each line-out terminal of the plurality of line-out terminals is in electrical communication with the circuit board.
- 16.** The initiator head of claim 13, further comprising: a ground terminal provided on a second side of the housing in the axial direction opposite to the first side; wherein the ground terminal is in electrical communication with the circuit board.
- 17.** The initiator head of claim 16, wherein: the ground terminal is one of a plurality of ground terminals; and each ground terminal of the plurality of ground terminals is in electrical communication with the circuit board.
- 18.** The initiator head of claim 13, further comprising a surface mounted component mounted on the circuit board.
- 19.** The initiator head of claim 13, wherein an exterior shape of the housing is rotationally asymmetric with respect to the axial direction.
- 20.** The initiator head of claim 19, further comprising a key protrusion extending radially outward from the housing.
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